

Logistic Analysis

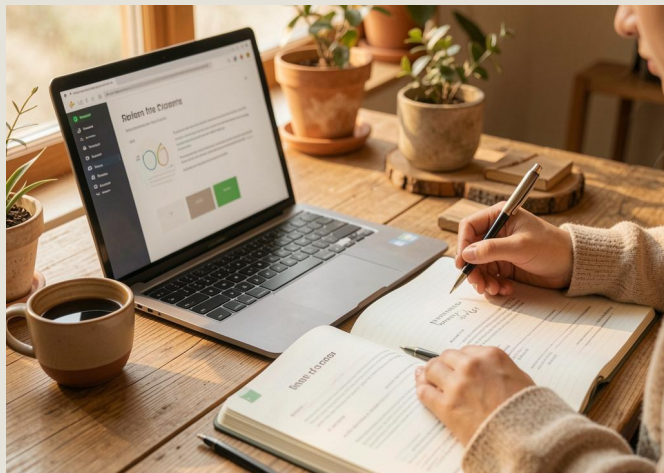
1

Sahin Cicek
Ethan Lee
Bhavaneeth Parnapalli

How do student
activities affect their
potential depression?

Introduction

2



Our dataset is a classification problem (meaning our output is 0 or 1) consisting of 18 columns with our response variable being the depression column. We removed three columns since they are not related to students, the focus of our project. Our data set consists of about 140 thousand data points.

01

Dataset Overview

Data was compiled from institutional surveys and lab records with defined inclusion criteria and collection period.

Documentation includes acquisition methods and initial validation to ensure reproducibility and provenance.

Featured students from various cities in India of all ages/studies, ranging from Class 12 to PhD students.

Data Source, Collection, and Characteristics

Sample characteristics

Overview of sample size, class balance, and key demographic or domain features.

Highlight representative vs. underrepresented groups and implications for model generalizability.



23 thousand observations remained after cleaning the data.

The regressors chosen after cleaning and assessing missing data patterns were Gender, Age, Academic Pressure, CGPA, Study Satisfaction, Study Hours, and Financial Stress.

Our plan was to apply appropriate *imputation* strategies and quality checks before modeling.

02

Data Preparation

As mentioned in the introduction, we dropped three unrelated columns for students. Since NA values can lead to errors in logistic regression, we removed them from our data. The only column containing NA's was "Financial Stress", so only that was data set was altered. We also removed cities that appear less than one hundred times to remove outliers.

```
df.isna().sum()
```

Degree	0
Work/Study Hours	0
Financial Stress	3
Illness	0
Depression	0

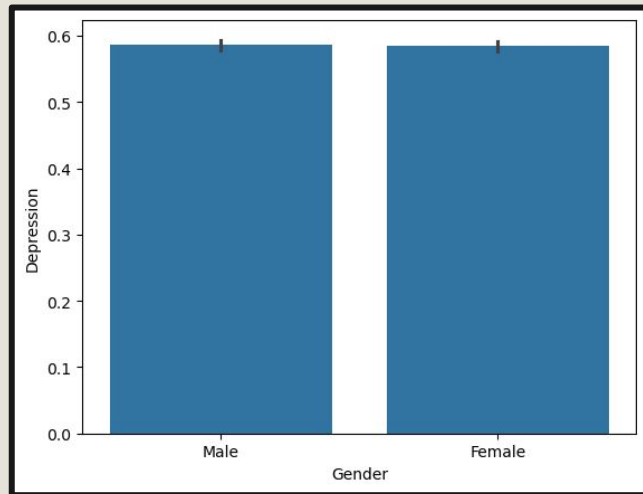
```
df = df[df['Profession'] == 'Student']
df = df.drop(columns=['Profession', 'Work Pressure', 'Job Satisfaction'])
df = df.dropna(subset=['Financial Stress'])
city_counts = df['City'].value_counts()
cities_to_group = city_counts[city_counts < 100].index
df['City'] = df['City'].replace(cities_to_group, 'Other')
```

Data
Preparation

Data Preparation

We wanted to see “Gender” column and its effect on depression. Surprisingly both male and female categories had the same ratio for depression. After running the Chi test and receiving p value of ~ 0.75 , we fitted the model with and without Gender. Since we got the same value on both, we decided to drop the entire column completely.

Note: This is not ideal in a real world scenario.



Chi-Square Statistic: 0.1004
p-value: 0.7513915658890091

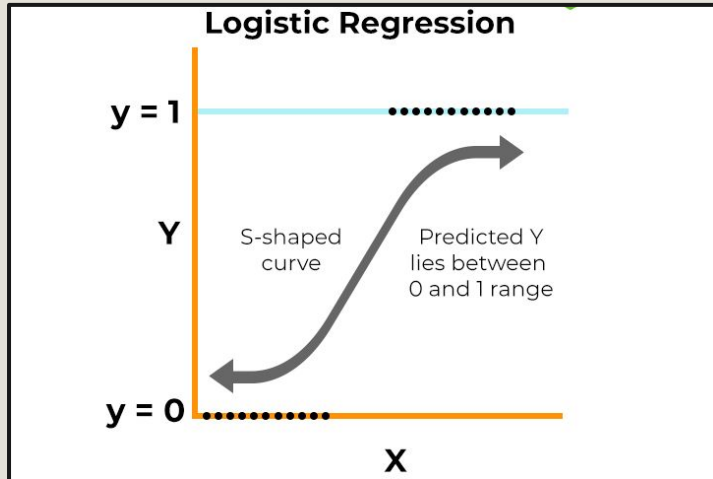
```
df = df.drop(columns=['Gender'])  
df = df.drop(columns=['City'])
```

Encoding & Selection

```
df_encoded = pd.get_dummies(df)
```

$$\frac{e^{a+bX}}{1 + e^{a+bX}}$$

Most machine learning models require numerical input, so categorical strings like 'BS' and 'PhD' must be converted into numerical form. We used "get_dummies" to assign each of these categories their own column with outputs 1 and 0 meaning yes and no.



Since our dataset is a classification problem, we opted to use a Logistic Regression.

Note:

Logistic regression calculates the probability of a given data point being in a class (i.g., Depression = 1). If it passess the threshold of 0.5, it is true; if not it is false.

03

Model Fitting

Model Fitting

=====							
Model	# Predictors	Log-Likelihood	AIC	BIC	Pseudo R ²	Test Accuracy	
Full	74	-7687.920306	15525.840611	16126.742712	0.492570	0.848583	
Reduced	43	-7721.660069	15531.320139	15883.849371	0.490343	0.848224	

Given the negligible difference in Pseudo R² and test accuracy, the 31 excluded predictors are not significant. Also, the reduced model having a lower BIC value means that the model fits better with less punishment, therefore we can continue using the reduced model.

FULL MODEL

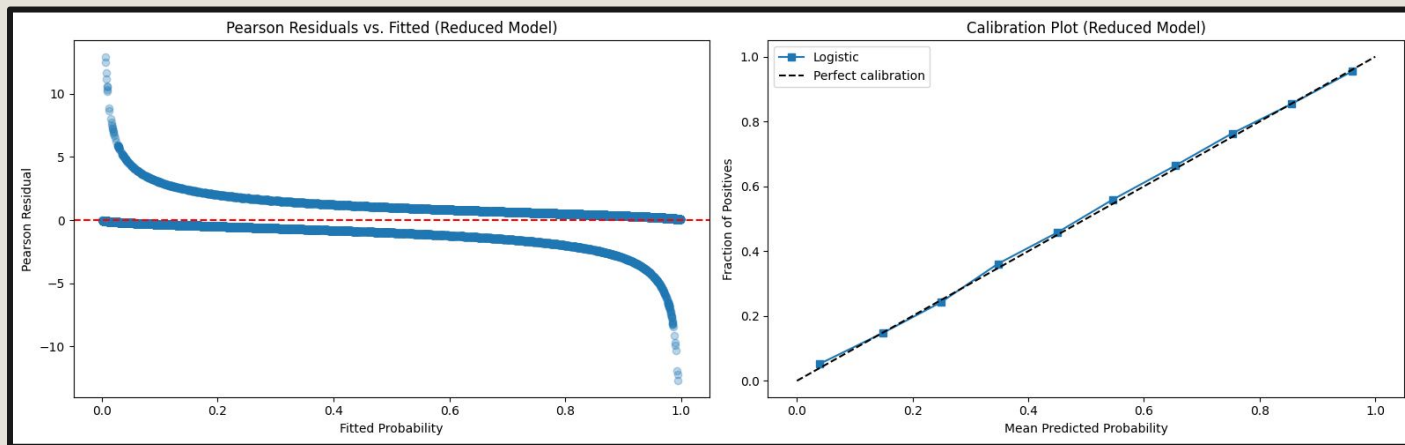
No. Observations:	22293
Df Residuals:	22218
Df Model:	74
Pseudo R-squ.:	0.4926
Log-Likelihood:	-7687.9
LL-Null:	-15151.
LLR p-value:	0.000

REDUCED MODEL

No. Observations:	22293
Df Residuals:	22249
Df Model:	43
Pseudo R-squ.:	0.4903
Log-Likelihood:	-7721.7
LL-Null:	-15151.
LLR p-value:	0.000

04

Residual Analysis

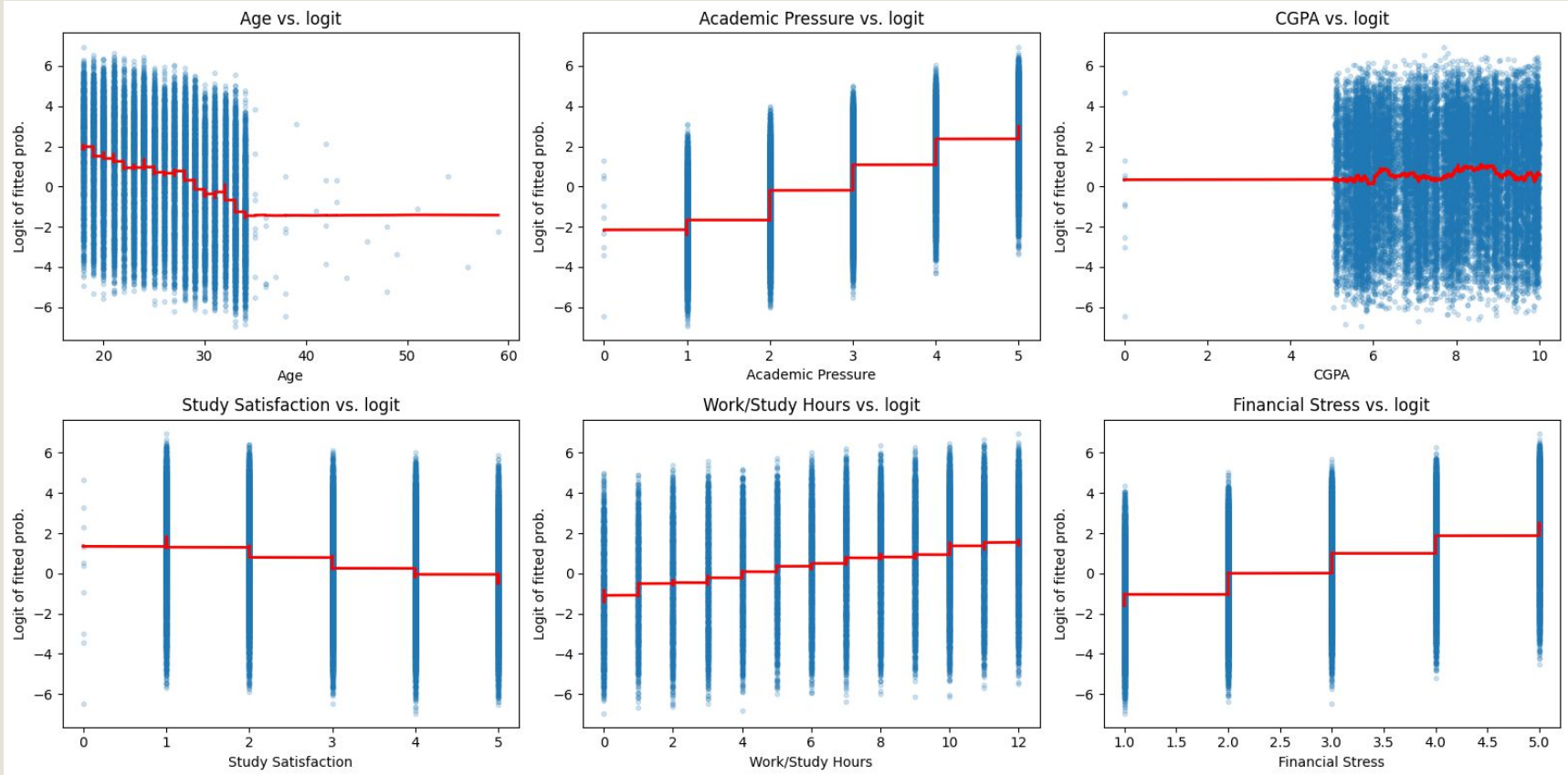


$$\frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}$$

The Pearson graph shows that most of the predictions are accurate because they fall within the $[-1, 1]$ range. The calibration plot confirms that the model's probabilities are reliable, meaning the predicted chances actually match real-world frequencies rather than being overconfident or biased.

Residual
Analysis

Do we need transformations? - Yes, outliers



Model Evaluation & Training

```
lr_X = df_encoded.drop(columns=['Depression'])
lr_y = df_encoded['Depression']
X_train, X_test, y_train, y_test = train_test_split(lr_X, lr_y,
                                                    test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000)
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
accuracy_score(y_test, y_pred)
```

✓ 0.1s

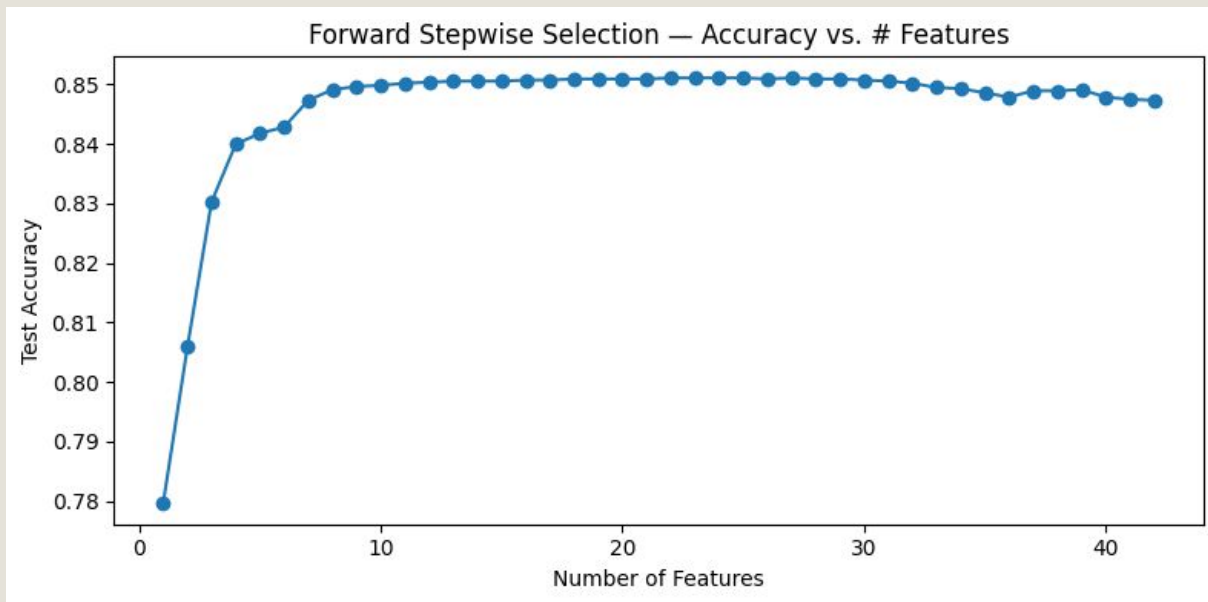
Python

0.8485827054180122

Because our prediction score is ~0.848, there is no need for further transformation.

05

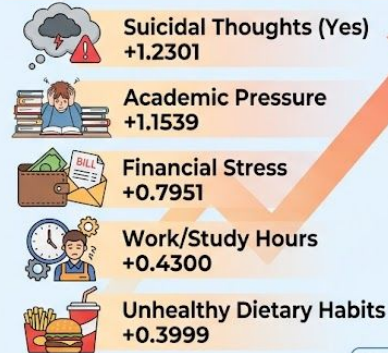
Selecting Model



We have several important features but other than them, our model overfits. We ran a final “forward Stepwise Selection” where we picked out only the features with reasonable impact, ending up with 22 features.

Logit Regression Analysis: Predictors of Student Depression Risk (Absolute Weight > 0.1)

FACTORS INCREASING RISK (Positive Coefficients)



FACTORS DECREASING RISK (Negative Coefficients)



Model Constant: 0.5662



```
logit(P(Depression=1)) = 0.5662 + 1.2301(Have you ever had suicidal thoughts ?_Yes) + 1.1539(Academic Pressure)
+ 0.7951(Financial Stress) - 0.5403(Age) - 0.3393(Study Satisfaction)
+ 0.4300(Work/Study Hours) + 0.3999(Dietary Habits_Unhealthy)
- 0.1656(Sleep Duration_More than 8 hours) - 0.0259(Degree_BA) + 0.0268(Degree_LLM)
- 0.0200(Degree_M.Tech) - 0.0345(Degree_MSc) + 0.0130(Degree_M.Ed)
- 0.0026(Sleep Duration_Others) - 0.0089(Degree_B.Ed) + 0.0195(Degree_Others)
- 0.0158(Degree_B.Pharm) - 0.0097(Degree_BCA) + 0.0056(Degree_BBA) + 0.0012(Degree_BE)
+ 0.0435(Degree_BSc) - 0.0207(Degree_MA)
```

Thank You!

Any
Questions?